Let F be the foot of the perpendicular from the point A to the plane $\vec{r}\cdot\vec{n}=p$. The line joining F and A is parallel to the normal vector $\vec{n}$ and hence its equation is $\vec{r}=\vec{u}+t\vec{n}$.

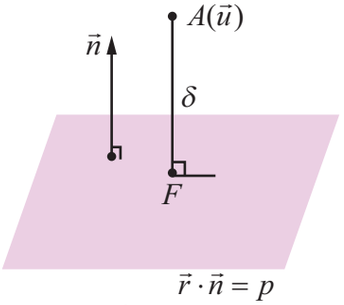


Fig. 6.32

But F is the point of intersection of the line $\vec{r}=\vec{u}+t\vec{n}$ and the given plane $\vec{r}\cdot\vec{n}=p$. If $\vec{r}_1$ is the position vector of F, then $\vec{r}_1=\vec{u}+t_1\vec{n}$ for some $t_1\in\mathbb{R}$, and $\vec{r}_1\cdot\vec{n}=p$ .Eliminating $\vec{r}_1$ we get

$(\vec{u}+t_1\vec{n})\cdot\vec{n}=p$ which implies $t_1=\dfrac{p-(\vec{u}\cdot\vec{n})}{|\vec{n}|^2}$.

Now, $$\overrightarrow{FA}=\vec{u}-(\vec{u}+t_1\vec{n}) \;=\; -t_1\vec{n}=\left(\frac{(\vec{u}\cdot\vec{n})-p}{|\vec{n}|^2}\right)\vec{n}$$

Therefore, the length of the perpendicular from the point A to the given plane is

$$\delta \;=\; \left|\overrightarrow{FA}\right|=\left|\left(\frac{(\vec{u}\cdot\vec{n})-p}{|\vec{n}|^2}\right)\vec{n}\right|=\left|\frac{(\vec{u}\cdot\vec{n})-p}{|\vec{n}|}\right|$$

The position vector of the foot F of the perpendicular AF is given by

$$\vec{r}_1 \;=\; \vec{u}+t_1\vec{n} \quad \text{or}$$

$$\vec{r}_1 \;=\; \vec{u}+\left(\frac{p-\vec{u}\cdot\vec{n}}{|\vec{n}|^2}\right)\vec{n}$$

Q2DRZW

(b) Equation of a plane in Cartesian form

In Caretesian form if $A(x_1,y_1,z_1)$ is the given point with position vector $\vec{u}$ and $ax+by+cz=p$ is the Cartesian equation of the given plane, then $\vec{u}=x_1\hat{i}+y_1\hat{j}+z_1\hat{k}$ and $\vec{n}=a\hat{i}+b\hat{j}+c\hat{k}$. Therefore, using these vectors in $\delta=\dfrac{|\vec{u}\cdot\vec{n}-p|}{|\vec{n}|}$, we get the perpendicular distance from a point to the plane in Cartesian form as

$$\delta=\left|\frac{ax_1+by_1+cz_1-p}{\sqrt{a^2+b^2+c^2}}\right|=\frac{|ax_1+by_1+cz_1-p|}{\sqrt{a^2+b^2+c^2}}$$

Remark

The perpendicular distance from the origin to the plane $ax+by+cz+d=0$ is given by

$$\delta=\frac{|d|}{\sqrt{a^2+b^2+c^2}}$$

Example 6.49

Find the distance of a point $(2,5,-3)$ from the plane $\vec{r}\cdot\left(6\hat{i}-3\hat{j}+2\hat{k}\right)=5$.

Solution

Comparing the given equation of the plane with $\vec{r}\cdot\vec{n}=p$, we have $\vec{n}=6\hat{i}-3\hat{j}+2\hat{k}$.

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